ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CE ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain 3-Phase Soil System. Define Void Ratio, Porosity, Degree of Saturation, Water Content. [14 Marks]

[OR]

Q.1 (Alt) Explain Index Properties: Liquid Limit, Plastic Limit, Plasticity Index, and Soil Classification (USCS/IS). [14 Marks]

--- UNIT II ---

Q.2 Explain Darcy's Law for soil permeability. Describe Constant Head and Falling Head permeameter tests. [14 Marks]

[OR]

Q.2 (Alt) Explain Effective Stress principle, Seepage Pressure, Quick Sand condition, and Flow Nets. [14 Marks]

--- UNIT III ---

Q.3 Explain Standard Proctor Compaction test. Compare Compaction vs Consolidation. [14 Marks]

[OR]

Q.3 (Alt) State Terzaghi's 1D Consolidation Theory. Calculate coefficient of consolidation (Cv) and settlement. [14 Marks]

--- UNIT IV ---

Q.4 Explain Shear Strength of soils: Mohr-Coulomb failure criterion. Describe Direct Shear Test. [14 Marks]

[OR]

Q.4 (Alt) Describe Triaxial Compression Test (UU, CU, CD tests) and Unconfined Compression Test. [14 Marks]

--- UNIT V ---

Q.5 Explain Terzaghi's Bearing Capacity Theory for shallow foundations with bearing capacity factors. [14 Marks]

[OR]

Q.5 (Alt) Explain Pile Foundations: Static and Dynamic pile driving formulas and group pile efficiency. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****